Ogbonna Arinze

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SUMMARY

Staff-caliber software engineer with 8 years of professional experience designing technical architecture, scaling software systems, and guiding engineering teams through complex decisions across healthcare, fintech, civic tech, and AI product environments. Proven track record architecting production platforms serving thousands of daily users, from monorepo-managed multi-application systems to real-time data streaming and AI-powered products. Combines deep hands-on fluency across frontend and backend technologies (React, Next.js, NestJS, TypeScript, PostgreSQL) with a consistent pattern of mentoring engineers, establishing organizational code quality standards, and driving cross-functional collaboration with product, design, and stakeholders.

CORE COMPETENCIES

- **System Architecture & Scalability:** Monorepos (Nx, Turborepo), SSR/SSG, micro-frontends, real-time streaming (SSE, EventSource), and performance optimization at scale.
- **Frontend & Backend:** React, Next.js, Vue.js, NestJS, NodeJS, TypeScript, Redux, Zustand, Tailwind CSS, Radix UI, and REST/GraphQL APIs.
- **AI & Intelligent Systems:** RAG pipelines, multi-agent architectures, OpenAI API, LangChain, Mastra, vector search (pgvector, Chroma), embeddings, and agentic workflows.
- **Databases & Infrastructure:** PostgreSQL, MongoDB, Prisma, Docker, AWS (CloudFront, S3), GitHub Actions, CI/CD pipelines, and production deployment orchestration.
- **Quality & Reliability:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Lighthouse, bundle analysis, and CI-driven quality gates.
- **Technical Leadership:** Architecture reviews, engineering standards, mentorship across experience levels, Agile/Scrum, TDD/BDD, and cross-functional stakeholder collaboration.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the technical architecture for the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, guiding the team through architectural trade-offs between web and extension paradigms while optimizing workflows for clinical site integrations.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, setting organizational standards for UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Optimized application performance and reliability by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, systematically identifying and resolving bottlenecks using **Lighthouse** and Chrome DevTools to reduce load times significantly.
- Implemented **JWT-based authentication** and **role-based access control** with HTTP-only cookies for secure token handling and permission-based UI rendering, collaborating closely with backend engineers to align on auth strategies.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components, improving developer experience and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence, code quality, and sustainable engineering practices.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of a scalable **React** frontend architecture, achieving a **25% boost in user engagement** and collaborating with software architects on system design decisions that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, improving speed and user experience at scale.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more reliable releases.

- Established organizational engineering standards for code quality using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers and guided frontend best practices, fostering a culture of high-quality, maintainable code and consistent design across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Designed and built a high-volume **bank Settlement Solution** in a team of 3, architecting the system to handle thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**, making complex architectural trade-offs for real-time transaction processing.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, evaluating and replacing socket.io for faster updates, lower overhead, and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** for streamlined user role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating with the business team to align technical solutions with product needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Designed and built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** similarity search + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking over 708K+ vector embeddings.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, conversation summarization, and memory extraction for a responsive, scalable user experience.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 projects), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing, **Langfuse** for LLM observability, and **Infisical** for secrets management.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*